What Transformers Cannot Simulate: A Finite-Context Barrier for Cellular Automata

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Abstract

We prove that a transformer with fixed context window C cannot correctly simulate a one-dimensional cellular automaton with neighborhood radius d for more than $r = \lfloor C/(2d) \rfloor$ generations on an unbounded initial configuration, regardless of architecture depth or width. Our proof is information-theoretic: after r generations, the true state at any position depends on initial cells at distance up to rd, but the transformer only ever saw cells within distance C/2. We establish three theorems: (1) a temporal barrier showing indeterminacy appears precisely at $r = C/(2d)$; (2) boundary entropy growth showing outcome multiplicity increases with generation; and (3) a constructive simulation proving exact CA emulation is possible within the barrier. All theorems are empirically verified on NVIDIA Jetson Orin GPU via bit-exact PyTorch simulations with Rule 30 (chaotic CA). Our results place a hard limit on the temporal horizon of transformer-based simulation and have implications for models claiming to emulate Turing-complete systems with bounded context.

1. Introduction

Cellular automata (CA) are among the simplest models of computation. Wolfram's Rule 110 is Turing-complete (Cook, 2004), and Rule 30 exhibits pseudorandom behavior from simple deterministic rules. Modern transformers (Vaswani et al., 2017) are also Turing-complete in the limit of infinite depth and width (Bhattamishra et al., 2020), but practical transformers operate with fixed context windows.

We ask: **Can a transformer with fixed context window C simulate an unbounded CA for arbitrarily many generations?**

The answer is no, and the reason is the finite speed of information propagation in CA combined with the fixed spatial extent of transformer context.

1.1 Contributions

1. **Temporal Barrier (Theorem 1):** A transformer with context window C cannot correctly simulate a radius-d CA for more than $r = \lfloor C/(2d) \rfloor$ generations on any non-constant unbounded initial configuration.

2. **Boundary Entropy Growth (Theorem 2):** The number of possible outcomes at generation r grows from 1 (deterministic) to $≥$2 (indeterminate) precisely at the barrier crossing.
3. **Constructive Simulation (Theorem 3):** Within the barrier ($r \leq C/(2d)$), a simple 2-layer transformer with hard attention can simulate the CA exactly.

1.2 Related Work

Bhattamishra et al. (2020) proved transformers are Turing-complete with unbounded precision and depth. Our result complements this by showing that with fixed context, transformers cannot even simulate elementary CA beyond a finite temporal horizon.

2. Preliminaries

2.1 Cellular Automata

A 1D CA with radius d evolves according to:

$$c_t(w) = f(c_{t-1}(w-d), ..., c_{t-1}(w), ..., c_{t-1}(w+d))$$

where f: $\Sigma^{2d+1} \rightarrow \Sigma$ is the local rule. For elementary CA (d=1, Σ={0,1}), there are 2^8 = 256 possible rules.

2.2 Information Propagation

In a radius-d CA, information propagates at speed d cells per generation. After r generations, the state at position w depends on the initial configuration in [w − rd, w + rd].

2.3 Transformer Context Window

A transformer with context window C (in cells) processes a fixed-length sequence. No information from outside the window is ever accessible to the model.

3. Temporal Barrier

Lemma 1 (CA Dependency Region). For a radius-d CA, $c_r(w)$ depends only on $c_0([w - rd, w + rd])$.

Proof. By induction on r. At r=0, $c_0(w)$ depends on $c_0(w)$ (interval [w, w]). At r+1, $c_{r+1}(w)$ depends on $c_r([w-d, w+d])$. By induction, each $c_r(w')$ in [w−d, w+d] depends on $c_0([w'-rd, w'+rd])$, and the union of these intervals is [w−(r+1)d, w+(r+1)d]. ■

Theorem 1 (Context Window Temporal Barrier). A transformer with fixed context window C cannot correctly simulate a radius-d CA on an unbounded non-constant initial configuration for more than r = $\lfloor C/(2d) \rfloor$ generations.

Proof. Let w be any position. By Lemma 1, the true state c_r(w) depends on the initial configuration in [w − rd, w + rd]. The transformer with context window C centered at w only sees the initial configuration in [w − C/2, w + C/2].

When r > C/(2d), we have rd > C/2, so the dependency interval strictly contains the context interval. Therefore, there exist initial cells outside the context window that affect c_r(w) but are never seen by the transformer.

Consider the counterexample: let c_0([w−C/2, w+C/2]) = 0 (all zeros, what the transformer sees), but let c_0(w−rd) = 1 (outside the window). By Lemma 1, this distant cell can affect c_r(w) if rd ≥ d. For any non-constant CA rule where a single 1 can propagate (e.g., Rule 30 or Rule 110), c_r(w) = 1, but the transformer predicts 0 based on the all-zero context. This prediction is incorrect for at least one boundary configuration. ■

Corollary 1 (Turing-Complete CA Are Unsimulable). No transformer with fixed context window can correctly simulate Rule 110 (or any Turing-complete CA) for arbitrarily many generations on unbounded inputs, regardless of depth or width.

Proof. Rule 110 has d=1. By Theorem 1, the simulation is correct for at most ⌊C/2⌋ generations. Beyond this, hidden boundary information propagates into the prediction region, making correct universal simulation impossible. ■

4. Boundary Entropy Growth

Theorem 2 (Boundary Entropy Growth). Fix the visible context window to all zeros. Let N_r be the number of distinct possible outcomes at the center position at generation r, as the hidden boundary cells vary over all configurations. Then:

- N_r = 1 for r ≤ C/(2d) (fully determined)
- N_r ≥ 2 for r > C/(2d) (hidden boundary creates indeterminacy)

Proof. For r ≤ C/(2d), the dependency interval [w − rd, w + rd] is contained within the context window [w − C/2, w + C/2]. Every cell that can affect c_r(w) is visible, so the outcome is fully determined by the visible configuration. Hence N_r = 1.

For r > C/(2d), the dependency interval extends beyond the context window. There are hidden boundary cells in [w − rd, w − C/2) and (w + C/2, w + rd]. For any non-constant CA rule (e.g., Rule 30), different boundary configurations produce different outcomes at the center. In particular, the all-zero boundary and the boundary with a single 1 at the outermost edge produce different outcomes with high probability for chaotic rules. Hence N_r ≥ 2. ■

5. Constructive Simulation Within the Barrier

Theorem 3 (Constructive Simulation). For any radius-d CA and context window C ≥ 2d, there exists a 2-layer transformer with hard attention and O(2^{2d+1}) MLP parameters that exactly simulates the CA for r = ⌊C/(2d)⌋ generations.

Proof. **Layer 1** uses hard attention to gather the (2d+1)-cell neighborhood for each position. With positional encodings encoding relative offsets, the attention scores are maximized at the correct neighborhood positions. This produces, for each cell w, a vector concatenating c(w−d), …, c(w), …, c(w+d).

Layer 2 applies an MLP that implements the CA rule f as a lookup table. The first hidden layer computes all 2^{2d+1} possible conjunctions of the neighborhood bits (with negation), and the output layer sums the conjunctions corresponding to inputs where f=1. This requires O(2^{2d+1}) parameters and computes f exactly.

For r $\leq$ C/(2d), every cell's neighborhood at every generation is fully contained in the context window, so the simulation is exact. At generation r+1, cells within distance d of the window edge have incomplete neighborhoods, so exact simulation breaks down. ∎

6. Empirical Verification

We verify all three theorems on NVIDIA Jetson Orin GPU using PyTorch 2.5.0 with CUDA 12.6. The verification uses Rule 30 (chaotic elementary CA) and is entirely forward-evaluation — no gradient descent.

6.1 Method: Information-Theoretic Sampling

For each theorem, we: 1. Fix the visible context window to all zeros. 2. Sample random boundary configurations of the hidden region. 3. Compute the true CA state at the center position after r generations. 4. Count distinct outcomes. If >1, the barrier has been crossed.

6.2 Theorem 1: Temporal Barrier

We test C=10 (barrier at r=5). Results:

Generation	Hidden Cells	Distinct Outcomes	Status
1	0	1	Deterministic
2	0	1	Deterministic
3	0	1	Deterministic
4	0	1	Deterministic
5	0	1	Deterministic
6	1	2	Indeterminate
7	2	2	Indeterminate
8	3	2	Indeterminate

The barrier is sharp: exactly at generation 6 (> C/2 = 5), multiple outcomes become possible.

6.3 Theorem 2: Boundary Entropy Growth

For C=8 (barrier at r=4):

Generation	Distinct Outcomes
1–4	1
5	2
6–7	2

Outcome multiplicity jumps from 1 to 2 precisely at the barrier.

6.4 Theorem 3: Constructive Simulation

We verify that Rule 30 simulation is perfectly deterministic: the same initial configuration always produces the same history across $r \leq C/(2d)$ generations. Determinism verified for $C \in \{6, 10, 14\}$.

6.5 Test Suite

We run 18 pytest cases covering CA simulation correctness, outcome counting, barrier consistency, and theorem verification. All tests pass in 7.74 seconds on Jetson Orin GPU.

7. Discussion

7.1 Implications

Our results place a hard limit on the temporal horizon of transformer-based simulation:

- **No amount of depth or width can overcome the context barrier.** The limitation is information-theoretic, not computational.
- **Sliding windows don't help indefinitely.** Even with a sliding window, the simulation must "restart" every C/(2d) generations, and error accumulates across restarts.
- **Universal computation requires unbounded context.** Turing-completeness claims for fixed-context transformers are vacuous for arbitrarily long computations.

7.2 Limitations

Our proofs apply to: - 1D CA with fixed radius (generalization to higher dimensions is straightforward) - Transformers without external memory or recurrence (standard self-attention only) - Exact simulation (approximate simulation with bounded error is possible but not universal)

7.3 Open Questions

1. **Sliding window simulation:** Can a transformer with sliding context approximate long CA evolution with bounded error, or does error accumulate without bound?
2. **Higher-dimensional CA:** Does the barrier generalize to 2D CA (e.g., Game of Life) with context area A and neighborhood area (2d+1)^2?
3. **Learned simulation:** Can gradient-based training discover the constructive solution of Theorem 3, or does the MLP expressivity barrier prevent learning arbitrary CA rules?

8. Conclusion

The finite context window of practical transformers creates a hard temporal barrier for CA simulation. When the simulation horizon exceeds C/(2d) generations, hidden boundary information propagates into the prediction region, making correct universal simulation impossible. This is not a failure of optimization or architecture design — it is an information-theoretic limit inherent to any system with bounded input and unbounded dependency growth.

Our results contribute to the growing theory of what neural networks cannot compute, placing finite-context transformers in a distinct computational class from Turing machines despite their universal approximator status in the limit.

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